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ORIGINAL ARTICLE

Effects of different levels of solar radiation and depth-varying vertical diffusion on the dynamics of a reaction–diffusion NPZ model

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Although the effects of solar radiation, diffusion and mixed layer depths on plankton populations have been studied extensively, it is not intuitive what the equilibrium states of a plankton system should be in the absence of seasonal forcing, nor what are the prime determiners of the attracting equilibrium states. In this study, the linear stability of a nutrient–phytoplankton–zooplankton (NPZ) model in reaction–diffusion form is analyzed to understand how solar radiation, diffusion and mixed layer depth influence the predicted equilibrium states. We incorporate parameters that control the response of the phytoplankton growth rate to solar radiation, control the depth of the upper mixed layer and control the strength of the diffusion. The combined effect of the constant in time parameters for these quantities is shown to significantly influence the stability properties of the NPZ reaction–diffusion model. In our parameter range, we find that diffusion and light are stronger determinants of the equilibrium state than mixed layer depth. This linear stability study gives insight into how these quantities affect the stability states of marine planktonic ecosystems and reinforces the importance of understanding light and upper mixed layer turbulence as important factors in determining the state of the plankton system.

KEYWORDS: NPZ models; plankton models; phytoplankton; dynamics; diffusion; mixed layer depth; solar radiation; stability

INTRODUCTION

Nutrient–phytoplankton–zooplankton (NPZ) models have been used to study marine plankton ecology, seasonal phytoplankton blooms and spatial plankton distribution among other natural processes (Franks, 2002; Kuhn *et al.*, 2015; Levy, 2015). However, the biological and physical processes that affect plankton communities are so numerous that no NPZ model can account for all of them explicitly and accurately. Many studies involving mathematical plankton models attempt to simulate additional biogeochemical processes by adding more state variables such as multiple species/groups of phytoplankton and/or zooplankton (Edwards, 2001; Gibson *et al.*, 2005) to accommodate for diversity in a plankton community. Some add a state variable for detritus (D) that simulates organic matter lost from plankton that must remineralize into nutrients (N) before it can be taken up again by phytoplankton. This introduces yet another parameter for the detritus remineralization rate, which makes it more difficult to understand the behavior of the model. Many such studies produce results that successfully mimic natural marine phenomena such as seasonal phytoplankton blooms, species composition of plankton communities in specific oceanic regions, and spatial plankton distribution through techniques such as parameter optimization (Kuhn *et al.*, 2015). However, interpreting the significance of the parameters from these complex models often comes from examining how well the simulated results match *in situ* records. Understanding the underlying dynamics of a model may bring more insight into its behavior and into the ecological principles governing plankton communities (Busenberg *et al.*, 1990; Edwards *et al.*, 2000).

Appropriate construction of an NPZ model depends on the nature and complexity of the process of interest. A system of coupled ordinary differential equations usually suffices for an NPZ model constructed for simulating biological processes without spatially dependent physical forcing. Busenberg *et al.* (Busenberg *et al.*, 1990) analyzed the dynamics of a simple, autonomous NPZ model composed of ordinary differential equations, giving insight into spatially independent marine plankton ecology. Their model and dynamical study has been used as a foundation for more complex models (Edwards *et al.*, 2000; Edwards, 2001). However, the ocean subjects plankton communities to physical forcing from solar radiation and from depth-dependent turbulent mixing, effects which are missing from this model.

Quantities such as solar radiation intensity, mixed layer thickness and turbulent diffusivity have been used to understand theoretical causes of phytoplankton blooms. These theories include, but are not limited to, the Critical

Depth Hypothesis (CDH) (Sverdrup, 1953), the Dilution Recoupling Hypothesis (DRH) (Behrenfeld, 2010) and the Critical Turbulence Hypothesis (CTH) (Huisman *et al.*, 1999). According to the CDH, which assumes a well-mixed surface layer, the depth of the mixed layer limits access to light, which limits growth. The CTH suggests that if turbulent mixing rates are less than growth rates, then a bloom will form. The DRH suggests that a deepening mixed layer depth reduces contact rates between the predators (zooplankton) and the prey (phytoplankton), allowing the phytoplankton population to flourish. Each of these theories has been invoked to explain observed patterns in plankton observations. For example, Nelson and Smith (Nelson and Smith, 1991) invoke the CDH to ice-edge phytoplankton blooms in the Southern Ocean. Taylor and Ferrari (Taylor and Ferrari, 2011) invoke the CTH as a foundation for their theory of spring phytoplankton blooms in the North Atlantic triggered by the spring shutdown of turbulent convection. Smith *et al.* (Smith *et al.*, 2015) invoke the DRH to explain North Atlantic phytoplankton blooms. However, *ad hoc* explanations that describe the temporal pattern in plankton populations may overlook important dynamical structures of the underlying plankton ecosystem.

Behrenfeld and Boss (Behrenfeld and Boss, 2014) suggest that changes in mixed layer depth and sunlight disrupts the balance of the phytoplankton–zooplankton equilibria. In other words, phytoplankton bloom formation or termination represent transitions from one ecological state to another as light and mixed layer depth change. Here, we incorporate the effects of solar radiation on growth rate, mixed layer depth and depth dependent diffusion into the ordinary differential equations NPZ model resulting in a nonlinear system of coupled partial differential equations. We do this to understand the impact of light, mixed layer depth and diffusion on the ecological stability of three equilibria: plankton-free, phytoplankton-only and coexisting phytoplankton and zooplankton.

To model the effect of solar radiation on phytoplankton growth, we use the linear growth response described by Franks (Franks, 2002), and we incorporate the effect of light attenuating with depth. Vertical mixing in the water column is commonly simulated with a diffusion term (Edwards *et al.*, 2000; Gibson *et al.*, 2005; Kuhn *et al.*, 2015; Levy, 2015). Diffusion is used as a proxy for turbulent mixing. The equilibria (and their stability) of a seasonally unforced NPZ reaction–diffusion system has been determined computationally by Edwards *et al.* (Edwards *et al.*, 2000). Here, we use the techniques from that study to analyze the effects of parameters representing solar radiation, vertical diffusion and mixed layer

Table I: Parameter descriptions and values for equations (1)

Parameter	Description	Value	Units
α	Assimilation efficiency	0.7	-
μ_{max}	Phytoplankton maximum growth rate	2	d^{-1}
a	Ivlev decay constant	1	$(\text{mmol N m}^{-3})^{-1}$
h_{max}	Maximum grazing rate	1.5	d^{-1}
k_N	Michaelis–Menten half-saturation constant	1	mmol N m^{-3}
m_P	Phytoplankton mortality rate	0.1	d^{-1}
m_Z	Zooplankton mortality rate	0.2	d^{-1}
I_0	Maximum seasonal surface intensity of solar radiation	30–60 ^a	$\text{E m}^{-2}\text{d}^{-1}$
N_T	Total nitrogen in the system	2	mmol N m^{-3}

Values from Gentleman and Neuheimer (Gentleman and Neuheimer, 2008). ^aValues vary geographically.

depth on the dynamics of a nonseasonally forced NPZ reaction–diffusion model.

This paper is organized as follows. First, we introduce a light response parameter to the model of Franks *et al.* (Franks *et al.*, 1986) and reinterpret its dynamical properties. We then increase the complexity of the model by attenuating light with depth and incorporating a depth dependent diffusion term. We linearize this system and compute the equilibria. In the section that follows, we present the results of the dynamical analysis with respect to light, diffusion and mixed layer depth. Finally, in the conclusions, we discuss practical implications of our results for NPZ models.

Ordinary differential equations model

We begin with the construction of a basic ordinary differential equation model. This is a system of three equations, one for each state variable: phytoplankton (P), zooplankton (Z) and nutrients (N). Assuming nitrogen is the limiting nutrient of system, the units of each state variable is mmol N m^{-3} .

$$\begin{aligned}
 \frac{dP}{dt} &= \frac{I}{I_0} \mu_{max} \frac{N}{k_N + N} P - h_{max}(1 - e^{-aP})Z - m_P P \\
 \frac{dZ}{dt} &= \alpha h_{max}(1 - e^{-aP})Z - m_Z Z \\
 \frac{dN}{dt} &= -\frac{I}{I_0} \mu_{max} \frac{N}{k_N + N} P + (1 - \alpha)h_{max}(1 - e^{-aP})Z \\
 &\quad + m_P P + m_Z Z
 \end{aligned} \tag{1}$$

The model is inspired by Franks *et al.* (Franks *et al.*, 1986), but here we add a linear light response, $\frac{I}{I_0}$ where $I = I(t)$ is the intensity of the solar radiation at the surface and I_0 is the assumed annual maximum of I , thus we have $0 < \frac{I}{I_0} \leq 1$. For simplicity, we henceforth refer to the quantity $\frac{I}{I_0}$ as the parameter ρ . The nutrient uptake rate of phytoplankton is described by the Michaelis–Menten function $g(N) = \mu_{max} \frac{N}{k_N + N}$. The

parameter α is the zooplankton’s assimilation efficiency of phytoplankton. The grazing rate of phytoplankton by zooplankton is represented by the Ivlev function $h(P) = (1 - e^{-aP})$. Table I gives a description of the parameters and our chosen values for them. Since the right-hand side of the system sums to zero, then $N + P + Z = N_T$, where the constant N_T is the total concentration of nitrogen in the system. Thus, we can reduce system (1) to the two equations (2).

$$\begin{aligned}
 \frac{dP}{dt} &= \rho \mu_{max} \frac{N_T - P - Z}{k_N + N_T - P - Z} P - \\
 &\quad h_{max}(1 - e^{-aP})Z - m_P P \\
 \frac{dZ}{dt} &= \alpha h_{max}(1 - e^{-aP})Z - m_Z Z
 \end{aligned} \tag{2}$$

A stability analysis of the coupled equations (2) without the parameter ρ was carried out by Busenberg *et al.* (Busenberg *et al.*, 1990). In their paper, there are three equilibria. The equivalent equilibria when the parameter ρ is included are:

- a “plankton-free” equilibrium $(P, Z, N) = (0, 0, N_T)$,
- a “phytoplankton-only” equilibrium $(P, Z, N) = (\bar{P}, 0, N_T - \bar{P})$,
- a “coexisting” equilibrium $(P, Z, N) = (P^*, Z^*, N_T - P^* - Z^*)$,

where

$$\bar{P} = N_T - \frac{m_P k_N}{\rho \mu_{max} - m_P}, P^* = -\frac{1}{a} \ln \left(1 - \frac{m_Z}{\alpha h_{max}} \right) > 0, \tag{3}$$

and Z^* being the unique solution, satisfying $0 < Z^* < N_T - P^*$, of the quadratic equation.

$$\begin{aligned}
 Z^2 - Z \left[\frac{\alpha (\rho \mu_{max} - m_P)}{m_Z} P^* + k_N + N_T - P^* \right] \\
 + P^* \frac{\alpha}{m_Z} [(\rho \mu_{max} - m_P)(N_T - P^*) - k_N m_P] = 0
 \end{aligned} \tag{4}$$

Busenberg *et al.* (Busenberg *et al.*, 1990) defined the following additional parameters. As noted previously, their analysis did not include the parameter ρ ; here we have extended their analysis to include it.

$$R_0 = \frac{\rho \mu_{\max} N_T}{m_P (k_N + N_T)} \quad (5)$$

$$R_1 = \frac{\alpha h_{\max} (1 - e^{-a\bar{P}})}{m_Z} \quad (6)$$

$$R_2 = \frac{\rho \mu_{\max} - m_P}{ah_{\max} \bar{Z}^* \left(1 - \frac{m_Z}{\alpha h_{\max}}\right) + \frac{\rho k_N \mu_{\max} (k_N + N_T - \bar{Z}^*)}{(k_N + N_T - \bar{Z}^* - P^*)^2}}. \quad (7)$$

These additional parameters determine the existence and stability of the equilibria according to the following conditions.

1. The equilibrium $(P, \bar{Z}, N) = (0, 0, N_T)$ always exists and is stable if and only if $R_0 < 1$.
2. If $R_0 > 1$ then the equilibrium $(\bar{P}, 0, N_T - \bar{P})$ exists and is stable if and only if $R_0 > 1$ and $R_1 < 1$.
3. If $R_1 > 1$ then the equilibrium $(P^*, \bar{Z}^*, N_T - P^* - \bar{Z}^*)$ exists and is stable if and only if $R_0 > 1$, $R_1 > 1$, and $R_2 < 1$.
4. If $R_2 > 1$ then $(P^*, \bar{Z}^*, N_T - P^* - \bar{Z}^*)$ is unstable but is surrounded by at least one closed orbit indicating a periodic solution.

Changing the parameter ρ , while holding the other model parameters constant, changes which of these stability conditions hold. We define the parameter

$$\bar{P}_E = \frac{m_P (k_N + N_T)}{\mu_{\max} N_T} \quad (8)$$

such that the phytoplankton-only equilibrium exists if $\rho > \bar{P}_E$ and we define

$$\bar{P}_S = \frac{m_P}{\mu_{\max}} \left[1 + \frac{k_N}{N_T - \frac{1}{a} \ln \left(\frac{\alpha h_{\max}}{\alpha h_{\max} - m_Z} \right)} \right] \quad (9)$$

such that the phytoplankton-only equilibrium is stable if $\rho < \bar{P}_S$. Then from (5), (6) and the second stability condition, we have existence and stability of the phytoplankton-only equilibrium when

$$\bar{P}_E < \rho < \bar{P}_S. \quad (10)$$

From (9), we see that $\bar{P}_S$ increases with m_Z and decreases with $h_{\max}$ and a . Thus, increasing m_Z or decreasing $h_{\max}$ or a causes the interval $(\bar{P}_E, \bar{P}_S)$ to widen, which increases the range of ρ for which the

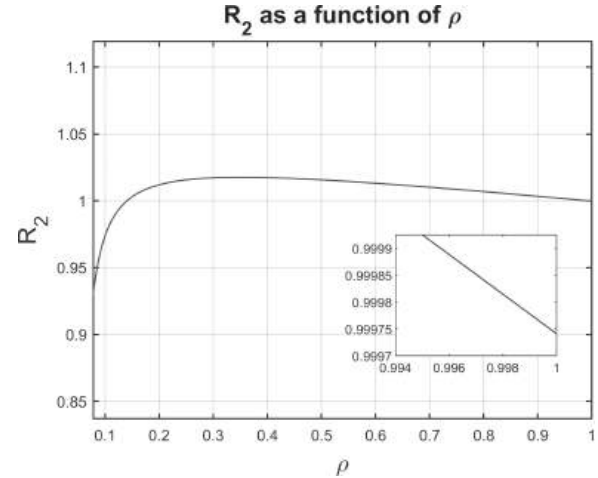


Fig. 1. R_2 plotted as a function of ρ . Only values of ρ where $\rho > \bar{P}_S$ are considered since the coexisting equilibrium does not exist where $\rho < \bar{P}_S$. Embedded graph shows that $R_2 < 1$ when ρ is close to 1. Parameter values from Table 1.

phytoplankton-only equilibrium is stable. When $\rho > \bar{P}_S$, the coexisting equilibrium is stable if $R_2 < 1$. With the parameter values from Table 1, we have $\bar{P}_E = 0.075$ and $\bar{P}_S = 0.078$. Note that R_2 is not a monotonic function of ρ ; see Fig. 1. There is no guarantee that $R_2 > 1$ just because ρ is greater than some value. The embedded graph in Fig. 1 shows that $R_2 < 1$ for all values of ρ in a small interval inclusive of 1 even though there are smaller values of ρ for which $R_2 > 1$. Figure 2 shows phase planes of the solution of (2) where the plankton-free equilibrium is stable ($\rho < \bar{P}_E$), where the phytoplankton-only equilibrium is stable ($\bar{P}_E < \rho < \bar{P}_S$), where the coexisting equilibrium is stable ($\rho > \bar{P}_S$) and ($R_2 < 1$), and where the coexisting equilibrium is unstable and repels nearby trajectories toward a closed orbit ($R_2 > 1$).

Reaction–diffusion model

Construction of the reaction–diffusion model

Now that a light reaction term is included, we expand equations (2) into a reaction–diffusion system (11) with depth ξ .

$$\begin{aligned} \partial_t P &= \rho e^{-k_W \xi} \mu_{\max} \frac{N_T - P - \bar{Z}}{k_N + N_T - P - \bar{Z}} P \\ &\quad - h_{\max} (1 - e^{-aP}) \bar{Z} - m_P P + \partial_\xi (D \partial_\xi P) \\ \partial_t \bar{Z} &= \alpha h_{\max} (1 - e^{-aP}) \bar{Z} - m_Z \bar{Z} + \partial_\xi (D \partial_\xi \bar{Z}) \\ D(\xi, t) &= D_{\min} - \frac{D_{\min} - D_{\max}}{2} \left(1 + \tanh \left(\frac{M(t) - \xi}{\psi} \right) \right) \end{aligned} \quad (11)$$

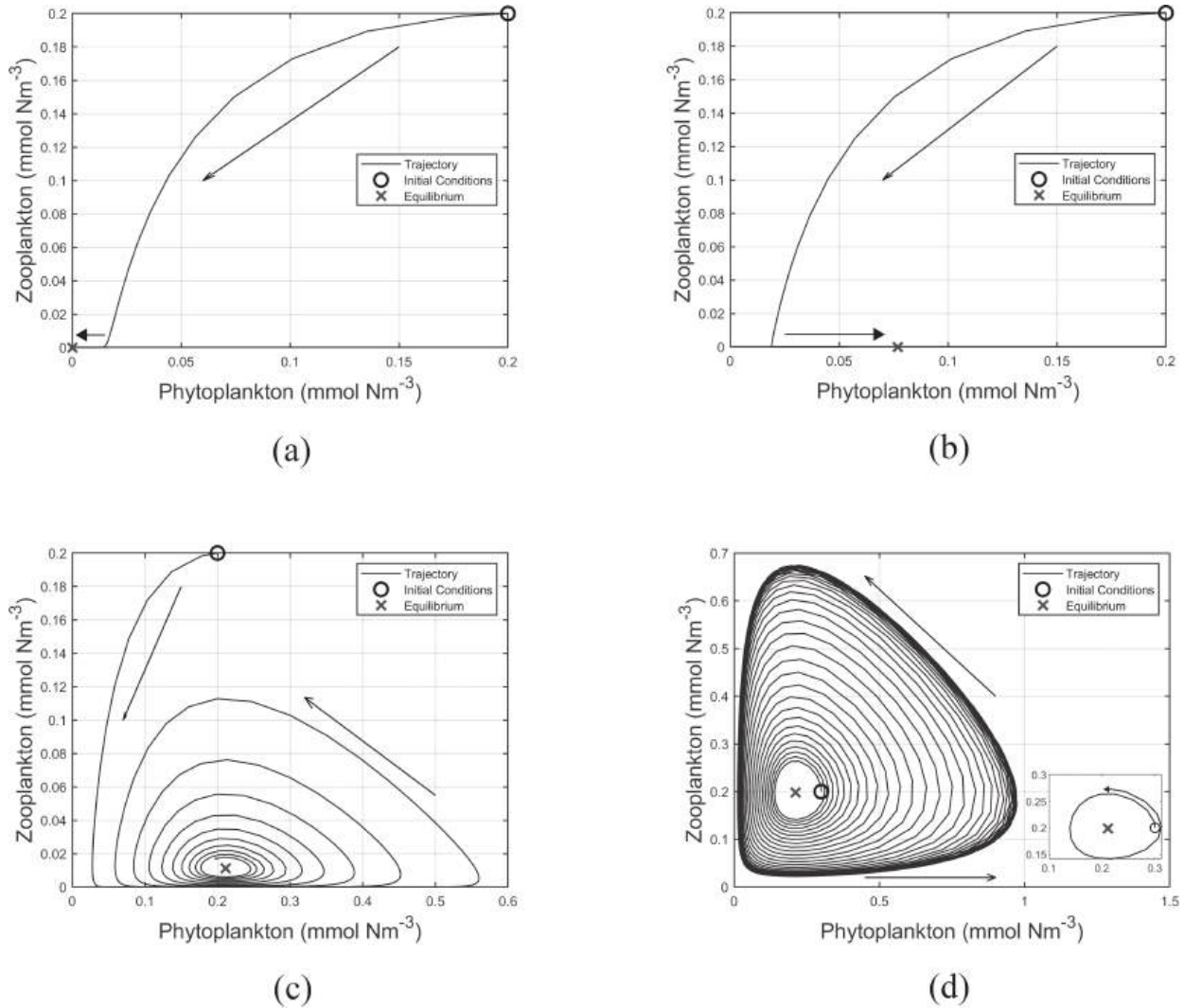


Fig. 2. Phase plane trajectories of solutions of equations (2) illustrating the three types of equilibria with parameter values from Table I. In each case, the trajectory starts at the circle. The stable equilibrium is marked by an “X” except in (d) where “X” is an unstable coexisting equilibrium. In (a), $\rho = 0.07$, implying $\rho < \bar{P}_E$ which means the plankton-free equilibrium is the only equilibrium and stable. In (b), $\rho = 0.076$, implying $\bar{P}_E < \rho < \bar{P}_S$ which means the phytoplankton-only equilibrium is the only attractor. In (c) $\rho = 0.09$, implying $\rho > \bar{P}_S$ and, from Fig. 1, $R_2 < 1$. This means that the coexisting equilibrium the only attractor. In (d) $\rho = 0.3$ which, from Figure 1, means $R_2 > 1$, indicating that all equilibria are unstable. A closed orbit surrounds the unstable coexisting equilibrium.

The phytoplankton response to light is now given by the expression $\rho e^{-k_W \xi}$, which accounts for light attenuating exponentially with depth. The diffusion coefficient $D(\xi, t)$, similar to that of Gibson *et al.* (Gibson *et al.*, 2005), resembles a two-layered vertical profile of the ocean. Franks (Franks, 2015) reiterates the important difference between turbulence and mixing. In this paper, however, we make a simplifying assumption by using diffusion as a proxy for mixing resulting from turbulence. The upper mixed layer has diffusion D_{max} , and the lower ocean has diffusion D_{min} . $M(t)$ is the mixed layer depth (i.e. the depth of the bottom of the upper mixed layer). The parameter ψ controls the thickness of the transition

layer between the mixed layer and the lower ocean. We choose $\psi = 25$ m for a smooth transition from the mixed layer to the lower ocean. This is computationally suitable for the vertical grid size of $\Delta \xi = 5$ m, which we use when we solve the system (11) in future research. This vertical grid size was also chosen by Levy (Levy, 2015) and Kuhn *et al.* (Kuhn *et al.*, 2015). Furthermore, experiments with $\psi = 5$ m, a more realistic choice, did not significantly change the results. Table II gives the descriptions of and the values chosen for the parameters we have introduced to equations (2) by expanding it into the reaction–diffusion system (11). Figure 3 shows a profile of the depth-dependent solar radiation intensity

Table II: Parameter descriptions and values introduced in equations (11)

Parameter	Description	Value	Units
k_W	Light attenuation constant ^a	0.06	m^{-1}
ψ	Transition layer parameter ^b	25	m
$D_{\min}$	Diffusion level in the lower ocean ^c	10^{-5}	$\text{m}^2 \text{s}^{-1}$
$D_{\max}$	Diffusion level in the upper mixed layer ^c	10^{-3}	$\text{m}^2 \text{s}^{-1}$
H	Water column depth	350	m

^aValue from Edwards *et al.* (Edwards *et al.*, 2000). ^bValue chosen for a smooth transition from mixed layer to lower ocean. ^cValue from Gibson *et al.* (Gibson *et al.*, 2005). Values for $D_{\min}$ and $D_{\max}$ are appropriate conditions for the Gulf of Alaska in spring. Values in other areas of the ocean can be $\mathcal{O}(2)$ larger or $\mathcal{O}(2)$ smaller

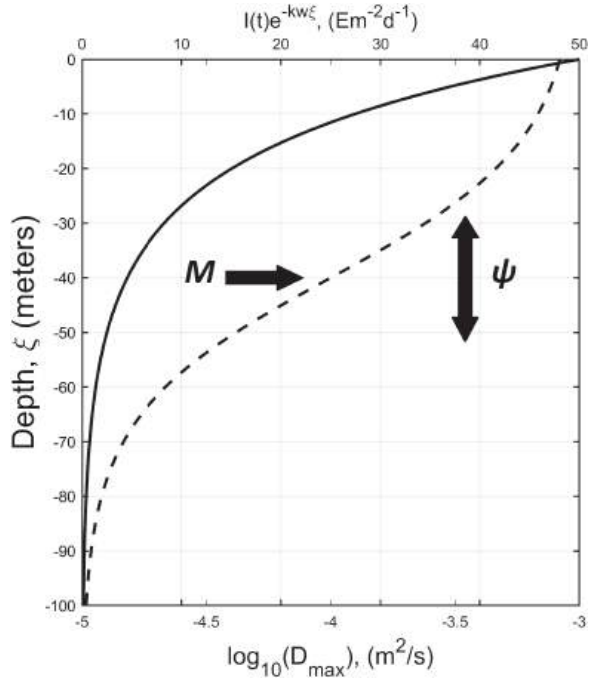


Fig. 3. Depth profile of solar radiation intensity (solid) and diffusion coefficient (dashed) with surface solar radiation intensity $I = 50 \text{ Em}^{-2}\text{d}^{-1}$ and mixed layer depth $M = 40 \text{ m}$ for purposes of illustration. Light attenuates exponentially with depth. The mixed layer depth M is at the inflection point of the diffusion profile and at the center of the transition layer. M varies seasonally and geographically and typically ranges from 20 to 300 meters.

and of the diffusion coefficient. The depth of the water column is set to 350 meters, as in Levy (Levy, 2015), to accommodate for most values of M . We impose no-flux boundary conditions at the surface and at the bottom. The system is solved using a Crank–Nicolson scheme.

Linearizing the reaction–diffusion system and finding the Equilibria

To begin the stability analysis of the reaction–diffusion system (11), the time dependence of ρ and M is initially

removed for them to be constants. Thus, we have.

$$\begin{aligned}
 \partial_t P &= \rho e^{-k_W \xi} \mu_{\max} \frac{N_T - P - \tilde{Z}}{k_N + N_T - P - \tilde{Z}} P \\
 &\quad - h_{\max} (1 - e^{-aP}) \tilde{Z} - m_P P + \partial_{\xi} (D \partial_{\xi} P) \\
 \partial_t \tilde{Z} &= \alpha h_{\max} (1 - e^{-aP}) \tilde{Z} - m_Z \tilde{Z} + \partial_{\xi} (D \partial_{\xi} \tilde{Z}) \\
 D(\xi) &= D_{\min} - \frac{D_{\min} - D_{\max}}{2} \left(1 + \tanh \left(\frac{M - \xi}{\psi} \right) \right)
 \end{aligned} \quad (12)$$

The system (12) is linearized using the technique from Edwards *et al.* (Edwards *et al.*, 2000). Let $(\tilde{P}(\xi), \tilde{Z}(\xi))$ be an equilibrium of (12) and $P'(\xi, t) = \hat{P}(\xi) e^{\lambda t}$ be a “perturbation” of the equilibrium such that $P'(\xi, t) = P(\xi, t) - \tilde{P}(\xi)$. If the system (12) is abbreviated as:

$$\begin{aligned}
 \partial_t P &= F(P, \tilde{Z}) + \partial_{\xi} (D \partial_{\xi} P) \\
 \partial_t \tilde{Z} &= G(P, \tilde{Z}) + \partial_{\xi} (D \partial_{\xi} \tilde{Z})
 \end{aligned} \quad (13)$$

then it can be written that:

$$\begin{aligned}
 \lambda \hat{P} e^{\lambda t} &= \partial_t P \\
 &= F(P, \tilde{Z}) + \partial_{\xi} (D \partial_{\xi} P) \\
 &= F(\tilde{P} + P', \tilde{Z} + \tilde{Z}') + \partial_{\xi} (D \partial_{\xi} (\tilde{P} + P')) \\
 &= F(\tilde{P}, \tilde{Z}) + P' \partial_P F(\tilde{P}, \tilde{Z}) + \tilde{Z}' \partial_{\tilde{Z}} F(\tilde{P}, \tilde{Z}) \\
 &\quad + \partial_{\xi} (D (\partial_{\xi} \tilde{P} + \partial_{\xi} P')) + \mathcal{O}((P')^2, (\tilde{Z}')^2, P' \tilde{Z}') \\
 &\approx F(\tilde{P}, \tilde{Z}) + \partial_{\xi} (D \partial_{\xi} \tilde{P}) + P' \partial_P F(\tilde{P}, \tilde{Z}) \\
 &\quad + \tilde{Z}' \partial_{\tilde{Z}} F(\tilde{P}, \tilde{Z}) + \partial_{\xi} (D \partial_{\xi} P') \\
 &= \hat{P}(\xi) e^{\lambda t} \partial_P F(\tilde{P}, \tilde{Z}) + \hat{Z}(\xi) e^{\lambda t} \partial_{\tilde{Z}} F(\tilde{P}, \tilde{Z}) \\
 &\quad + e^{\lambda t} \partial_{\xi} (D \partial_{\xi} \hat{P})
 \end{aligned} \quad (14)$$

Therefore,

$$\partial_P F(\tilde{P}, \tilde{Z}) \hat{P} + \partial_{\tilde{Z}} F(\tilde{P}, \tilde{Z}) \hat{Z} + \partial_{\xi} (D \partial_{\xi} \hat{P}) = \lambda \hat{P}, \quad (15)$$

and similarly,

$$\partial_P G(\tilde{P}, \tilde{Z}) \hat{P} + \partial_{\tilde{Z}} G(\tilde{P}, \tilde{Z}) \hat{Z} + \partial_{\xi} (D \partial_{\xi} \hat{Z}) = \lambda \hat{Z}. \quad (16)$$

Equations (15) and (16) can be written as a matrix equation:

$$\begin{bmatrix} \frac{\partial F}{\partial P} \Big|_{(\tilde{P}, \tilde{Z})} + \frac{\partial}{\partial \xi} \left(D \frac{\partial}{\partial \xi} \right) & \frac{\partial F}{\partial Z} \Big|_{(\tilde{P}, \tilde{Z})} \\ \frac{\partial G}{\partial P} \Big|_{(\tilde{P}, \tilde{Z})} & \frac{\partial G}{\partial Z} \Big|_{(\tilde{P}, \tilde{Z})} + \frac{\partial}{\partial \xi} \left(D \frac{\partial}{\partial \xi} \right) \end{bmatrix} \begin{bmatrix} \hat{P} \\ \hat{Z} \end{bmatrix} = \lambda \begin{bmatrix} \hat{P} \\ \hat{Z} \end{bmatrix} \quad (17)$$

The stability of the equilibrium calculated depends on the eigenvalues λ of equation (17). Since we calculate the equilibria numerically (see below), then we must discretize the depth of the water column (i.e. choose a finite number of depth values ξ for which to approximate the value of the equilibria). If m depths are chosen, then the matrix in equation (17) will have size $2m \times 2m$. Therefore, there will be a total of $2m$ eigenvalues λ . If the real part of every eigenvalue λ for an equilibrium $(\tilde{P}, \tilde{Z})$ is negative, then the equilibrium is asymptotically stable. If even one eigenvalue has a positive real part, then the equilibrium is unstable. The linear stability of an equilibrium can quickly be determined by checking whether the largest real part of the eigenvalues is positive ($\max_{\lambda} \Re(\lambda) > 0$) or negative ($\max_{\lambda} \Re(\lambda) < 0$).

The depth-dependent equilibria of equations (12) are computed using the technique from Edwards *et al.* (Edwards *et al.*, 2000). After discretizing depth, we calculate the equilibrium (P, Z) , for constants P and Z , at each depth with $D \equiv 0$ (i.e. for each ξ we analytically calculate the equilibria for (18)).

$$\begin{aligned} \frac{dP}{dt} &= \rho e^{-k_W \xi} \mu_{\max} \frac{N_T - P - Z}{k_N + N_T - P - Z} P \\ &\quad - h_{\max} (1 - e^{-aP}) Z - m_P P \\ \frac{dZ}{dt} &= \alpha h_{\max} (1 - e^{-aP}) Z - m_Z Z \end{aligned} \quad (18)$$

Then we create vectors $\vec{P}$ and $\vec{Z}$ containing the values of P and Z , respectively, for each depth ξ . Using $(\vec{P}, \vec{Z})$ as an initial guess, we use a nonlinear equation solver to solve the system (19).

$$\begin{aligned} F(P, Z) + \partial_{\xi} (D \partial_{\xi} P) &= 0 \\ G(P, Z) + \partial_{\xi} (D \partial_{\xi} Z) &= 0 \end{aligned} \quad (19)$$

Figure 4 shows the depth profiles of the initial guess and the computed coexisting equilibrium of (12) using $\rho = 1$ and a mixed layer depth of 100 m.

RESULTS AND DISCUSSION

We investigate the effects of the parameters ρ (response to solar radiation), M (mixed layer depth) and $D_{\max}$ (mixed layer diffusion) on the equilibria of the system (12), which is calculated using the technique from the previous section. Stability diagrams (Fig. 5) are constructed such that ρ is plotted against either M or $D_{\max}$. All other parameters assume the values listed in Tables I and II. The white region is where the plankton-free equilibrium is stable. The dark gray region is where the phytoplankton-only equilibrium is stable. The light gray region is where the coexisting equilibrium is stable. No regions were found where the coexisting equilibrium is unstable (Fig. 2d). Edwards *et al.* (Edwards *et al.*, 2000) found a large region where the coexisting equilibrium is unstable when plotting $h_{\max}$ against diffusion D , where D is constant in time and depth. However, most of the region of instability for the coexisting equilibrium in their study is in areas where $h_{\max} > 1.5 \text{ d}^{-1}$ and $D < 10^{-5} \text{ m}^2 \text{ s}^{-1}$, which is outside the range of these parameters used in this paper. Also, we have observed that the introduction of a depth dependent diffusion coefficient in this model is associated with the absence of unstable coexisting equilibria.

The construction of these diagrams begins with computing the largest real part of the eigenvalues λ from equation (17) for each pair of values ρ and M (or ρ and $D_{\max}$) with $(\vec{P}, \vec{Z})$ taken as the plankton-free equilibrium. The borderline between the white region and the dark gray region represents the border between the plankton-free equilibria in which $\max_{\lambda} \Re(\lambda) < 0$ (stable) and the plankton-free equilibria in which $\max_{\lambda} \Re(\lambda) > 0$ (unstable). Next, $(\vec{P}, \vec{Z})$ is taken as the phytoplankton-only equilibrium. The borderline between the dark gray region and the light gray region represents the border between the phytoplankton-only equilibria in which $\max_{\lambda} \Re(\lambda) < 0$ (stable) and the phytoplankton-only equilibria in which $\max_{\lambda} \Re(\lambda) > 0$ (unstable). Separate stability diagrams for the coexisting equilibrium (not shown) were constructed. No regions of instability (i.e. behavior like in Fig. 2d) were found.

Figure 5 illustrates the importance of the parameters for light response ρ , mixed layer depth M and mixed layer diffusion $D_{\max}$ for understanding this model. Figure 5a, c, e, show ρ plotted against M for $D_{\max}$ values 10^{-2} , 10^{-3} and $10^{-4} \text{ m}^2 \text{ s}^{-1}$, respectively. The influence of the mixed layer depth on the stability of the equilibria increases with mixed layer diffusion. Mixed layer depth has less influence on the selection of the stable state when $D_{\max} = 10^{-4} \text{ m}^2 \text{ s}^{-1}$ except when M is small (Fig. 5e) but shows greater influence when $D_{\max} = 10^{-2} \text{ m}^2 \text{ s}^{-1}$ (Fig. 5a). Evidently, mixed layer depth has little influence

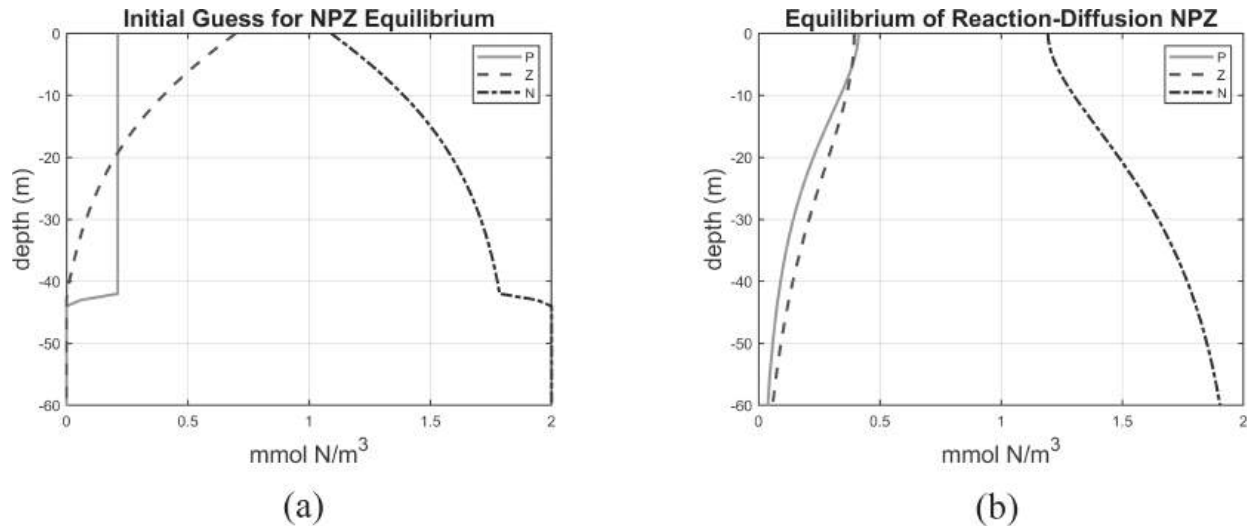


Fig. 4. Depth profiles of (a) the initial guess of the coexisting equilibrium of the reaction diffusion system (12) and (b) the computed coexisting equilibrium, i.e. the numerical solution of (19) with $\rho = 1$ and mixed layer depth set to 100 m.

except if it is very small or if D_{max} is quite large. However, Fig. 5b, d, f show that both ρ and D_{max} significantly influence which of the three equilibria is stable, regardless of the mixed layer depth.

The dashed lines represent seasonal increases of light response for moderate levels of mixed layer depth (or mixed layer diffusion) as winter progresses into summer. For each trajectory, the left end of the dashed line represents the winter solstice (when the solar radiation is at, say, 20% of the assumed seasonal maximum, which is typical in the North Atlantic, 40–45°N). The right arrow on the right end of the dashed line represents the summer solstice. In Fig. 5b, d, f, we see that when $D_{max} = 10^{-2}$, the phytoplankton-only equilibrium gains its stability at a higher light response value than it does when $D_{max} = 10^{-3}$. In Fig. 5e, we see that the coexisting equilibrium remains stable year-round, regardless of the mixed layer depth. In Fig. 5a, c, we see that when $M = 150$ m, the phytoplankton-only equilibrium gains its stability at a higher light response value than it does when $M = 25$ m. However, when $D_{max} = 10^{-3}$ (Fig. 5c), the phytoplankton-only equilibrium gains its stability at the same light response value for any mixed layer depth deeper than 100 meters.

The stability diagrams in Fig. 5 can indicate which ecological theories are credible for planktonic ecosystems with certain parameter regimes. For instance, one may consider using equations (12) to simulate seasonal phytoplankton blooms. As discussed in the introduction, Behrenfeld and Boss (Behrenfeld and Boss, 2014) suggest that blooms represent a change from one ecological state to another. Figure 5 shows that the model (12) can change

between three states (stable plankton-free equilibrium, stable phytoplankton-only equilibrium or stable coexisting equilibrium) as light response, mixed layer depth and mixed layer diffusion change. Consider the equations (12) with seasonally dependent light response ρ and mixed layer depth M , and constant mixed layer diffusion D_{max} . Choosing a high value of mixed layer diffusion such as $D_{max} = 10^{-2} \text{ m}^2 \text{ d}^{-1}$ causes the mixed layer depth to have a significant influence on which equilibrium is the attractor, as shown in Fig. 5a. A key assumption of Sverdrup's CDH (Sverdrup, 1953) is that the mixed layer be characterized by high turbulence. If this is the case, then Sverdrup suggests that changing mixed layer depth is the primary driving force behind the annual North Atlantic spring phytoplankton bloom. However, choosing a low value of mixed layer diffusion, such as $D_{max} = 10^{-4} \text{ m}^2 \text{ d}^{-1}$ (Fig. 5e) or $D_{max} = 10^{-3} \text{ m}^2 \text{ d}^{-1}$ (Fig. 5c), causes the mixed layer depth to have minimal (if any) effect on which equilibrium is attracting. Similarly, we suggest that only at high levels of diffusion does the mixed layer depth strongly influence which stable state of the ecosystem attracts. The mechanisms attracting plankton populations that occur in low diffusion environments are congruent with the CTH (Huisman *et al.*, 1999). According to this theory, if the ratio of mixed layer diffusion rate and growth rate is low, a phytoplankton population will be maintained, regardless of the mixed layer depth. Thus, the solution will be attracted to stable states with phytoplankton, which are the light gray and dark gray regions in Fig. 5.

Figure 6 shows three stability diagrams with ρ plotted against M and three stability diagrams with ρ plotted

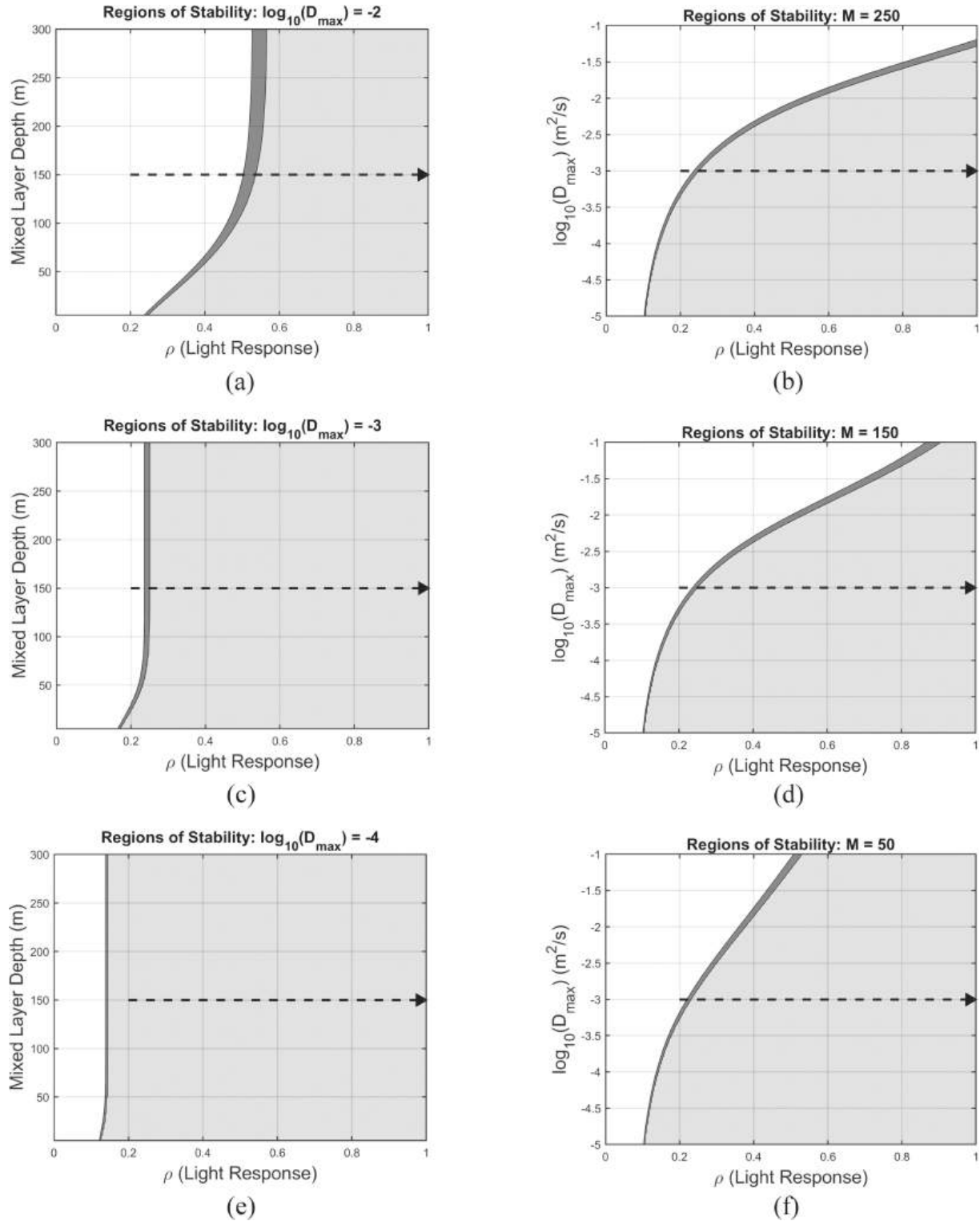


Fig. 5. Assuming parameter values from Tables I and II except where otherwise labeled, stability diagrams of equations (12). In (a), (c) and (e), light response is plotted against mixed layer depth, and in (b), (d) and (f), light response is plotted against mixed layer diffusion. White region: plankton-free equilibrium is stable (left region); dark gray region: phytoplankton-only equilibrium is stable (middle region); light gray region: coexisting equilibrium is stable (right region). Dashed lines represent seasonal increases of light response, with the summer solstice at right arrow, for a certain level of mixed layer depth or mixed layer diffusion.

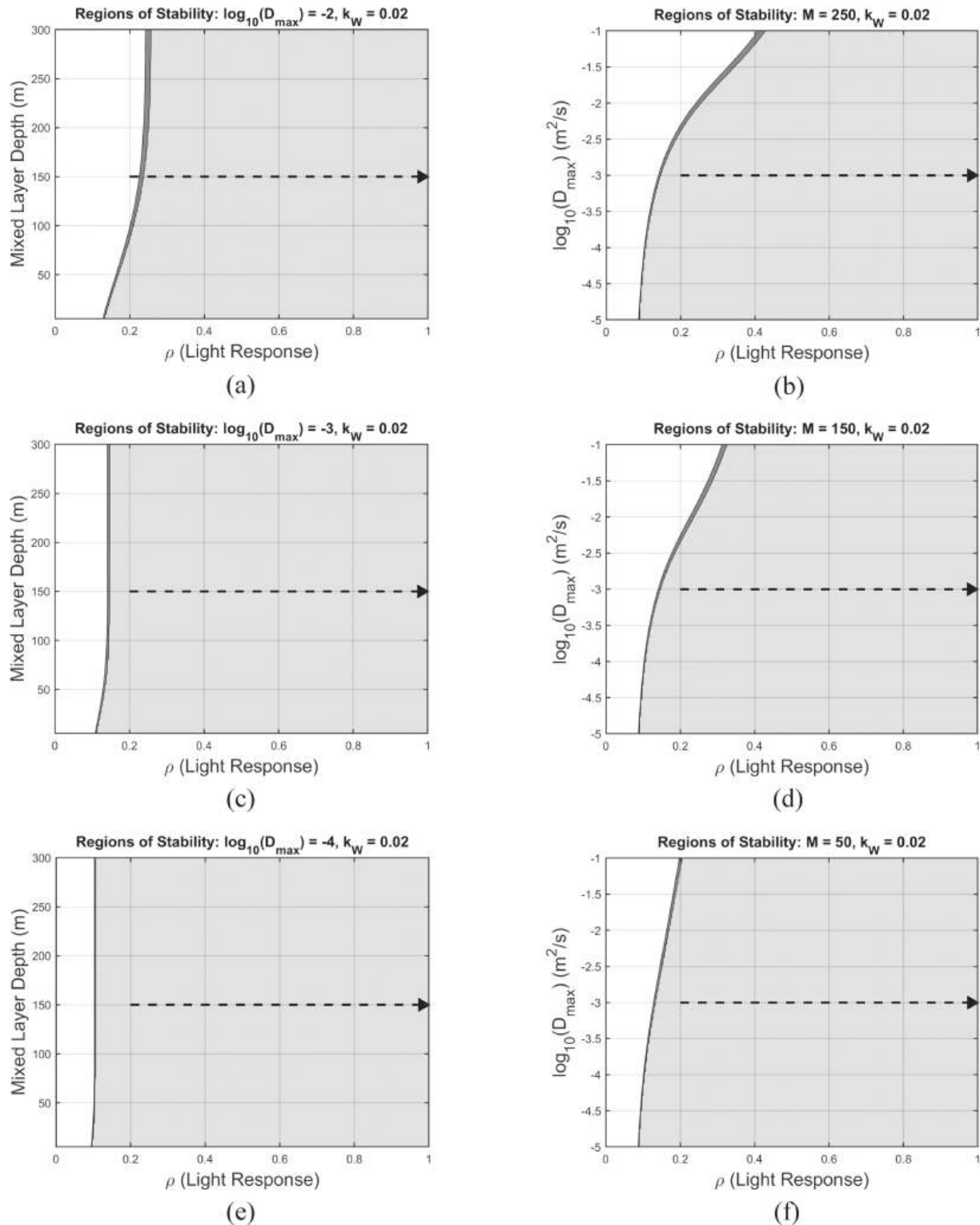


Fig. 6. Stability diagrams of equations (12) with parameter values from Tables I and II except where otherwise labeled. The diagrams are equivalent to those of Fig. 5 except that k_W is reduced from 0.06 to 0.02 m^{-1} . In (a), (c) and (e), light response is plotted against mixed layer depth, and in (b), (d) and (f), light response is plotted against mixed layer diffusion. White region: plankton-free equilibrium is stable (left region); dark gray region: phytoplankton-only equilibrium is stable (middle region); light gray region: coexisting equilibrium is stable (right region). Dashed lines represent seasonal increases of light response, with the summer solstice at right arrow, for a certain level of mixed layer depth or mixed layer diffusion.

against D_{max} as in Fig. 5 but with the light attenuation coefficient k_W reduced from 0.06 to 0.02 m^{-1} . These values approximate the seasonal range of k_W in the North Atlantic open ocean. This increases the light availability in the water column which reduces the area of the stable plankton-free equilibrium and increases the area of the stable coexisting equilibrium. This is expected because in equation (12) growth rate is linearly related to light availability. Therefore, increased light availability boosts the phytoplankton growth rate. This in turn increases the phytoplankton population, providing more prey for zooplankton, which facilitates the stability of the coexisting equilibrium. Reducing k_W “brightens” the ocean, roughly translating the regions of stability toward the left. Recall that the parameter values in Table I cause the phytoplankton-only equilibrium of equations (2) to be stable for $0.075 < \rho < 0.078$. Figs. 6e and 6f show that for low values of M , D_{max} and k_W , the phytoplankton-only equilibrium of equations (12) is stable when ρ is within a narrow range of values close to 0.08. Thus, for low values of mixed layer depth, diffusion and light attenuation, we expect similar dynamical behavior between the ordinary differential equations (2) and the reaction–diffusion system (12). The exception to this is that, for the system (12), we do not expect high values of ρ to be associated with unstable coexisting equilibria surrounded by periodic orbits.

Figure 7 shows three stability diagrams with ρ plotted against M and three stability diagrams with ρ plotted against D_{max} as in Fig. 5 but with the maximum grazing rate h_{max} reduced from 1.5 to 0.5 d^{-1} , the value chosen by Busenberg *et al.* (Busenberg *et al.*, 1990). This enlarges the region of stability for the phytoplankton-only equilibrium as we expect from the discussion of the $\bar{P}_S$ parameter in the discussion of our ordinary differential equations model. Consequently, reducing the maximum grazing rate h_{max} causes the phytoplankton-only equilibrium to be stable for a higher value of light response ρ . As can be seen from the second equation in equations (1), a lower value of h_{max} requires a higher phytoplankton concentration P for the zooplankton growth term $\alpha h_{max}(1 - e^{-aP})Z$ to be larger than the zooplankton loss term $m_Z Z$. In other words, a system with a lower maximum grazing rate requires a larger phytoplankton population (facilitated by increased sunlight) to sustain the zooplankton population.

The stability diagrams in Fig. 7 offer further insight from the discussion of Fig. 5. Again, consider equations (12) with seasonally dependent light response ρ and mixed layer depth M (or mixed layer diffusion D_{max}). Imagine now tracing increasing seasonal light along the arrow in Fig. 7b, when the mixed layer is deep. The widening of the phytoplankton-only stability region forces this equi-

librium to remain stable over more light values, which is a proxy for time in a seasonal cycle. Revisiting Behrenfeld and Boss’ (Behrenfeld and Boss, 2014) suggestion that blooms result from transitioning from one attracting ecological equilibrium to another, we suggest that Figure 7 shows that decreased grazing rates lead to a larger region of stability for the phytoplankton-only states. We believe this is analogous to the dilution-recoupling hypothesis (Behrenfeld, 2010), where increased mixed layer depth and vertical diffusion reduces the contact rates and therefore grazing rates between zooplankton and phytoplankton. Lower grazing rates increase the area where the phytoplankton-only equilibrium is an attractor, therefore leading to a bloom. Likewise, Behrenfeld and Boss (Behrenfeld and Boss, 2014) suggest that high-latitude blooms must follow from winter decimation of zooplankton, thus lowering grazing rates. This decimation might be the result of the ecosystem briefly entering a state in the winter where the phytoplankton-only equilibrium is attracting. The ecosystem then quickly returns to a state where the coexisting equilibrium is attracting so that the zooplankton population does not decline to a point from which it cannot recover. Although the equilibria influence the ecosystem, seasonally changing environmental conditions prevent the system from achieving any of the equilibrium states as explained by Hutchinson (Hutchinson, 1961). The region of stability for the phytoplankton-only equilibrium is very narrow for high values of maximum grazing rate h_{max} as seen in Fig. 5 and Fig. 6. Ecosystems with parameterizations like that of Fig. 5 or Fig. 6 would not tend toward a phytoplankton-only equilibrium for long. Plankton populations in sufficiently low sunlight and high mixed layer diffusion (or deep mixed layer depth) would no longer be attracted by a stable coexisting equilibrium but be attracted by a plankton-free state only after a brief attraction by a phytoplankton-only equilibrium. Intuitively, these are not ideal conditions for facilitating a phytoplankton bloom. Therefore, it is appropriate that equations (12) be parameterized so that the model has a sufficiently large region of stability for the phytoplankton-only equilibrium.

The simplicity of the reaction–diffusion model in this paper calls for caution when interpreting its results. The model does not account for sinking of phytoplankton or detritus, which occurs in nature. The model also does not account for the deepening of the mixed layer causing the entrainment of nutrients from the depths into the photic zone. The assumption that $N + P + Z = N_T$ for constant N_T implies that no nutrients leave the upper ocean, where the biological interactions between nutrients, phytoplankton and zooplankton occur. If sinking of phytoplankton (or detritus) and entrainment of nutrients into the photic zone were to be incorporated, then the

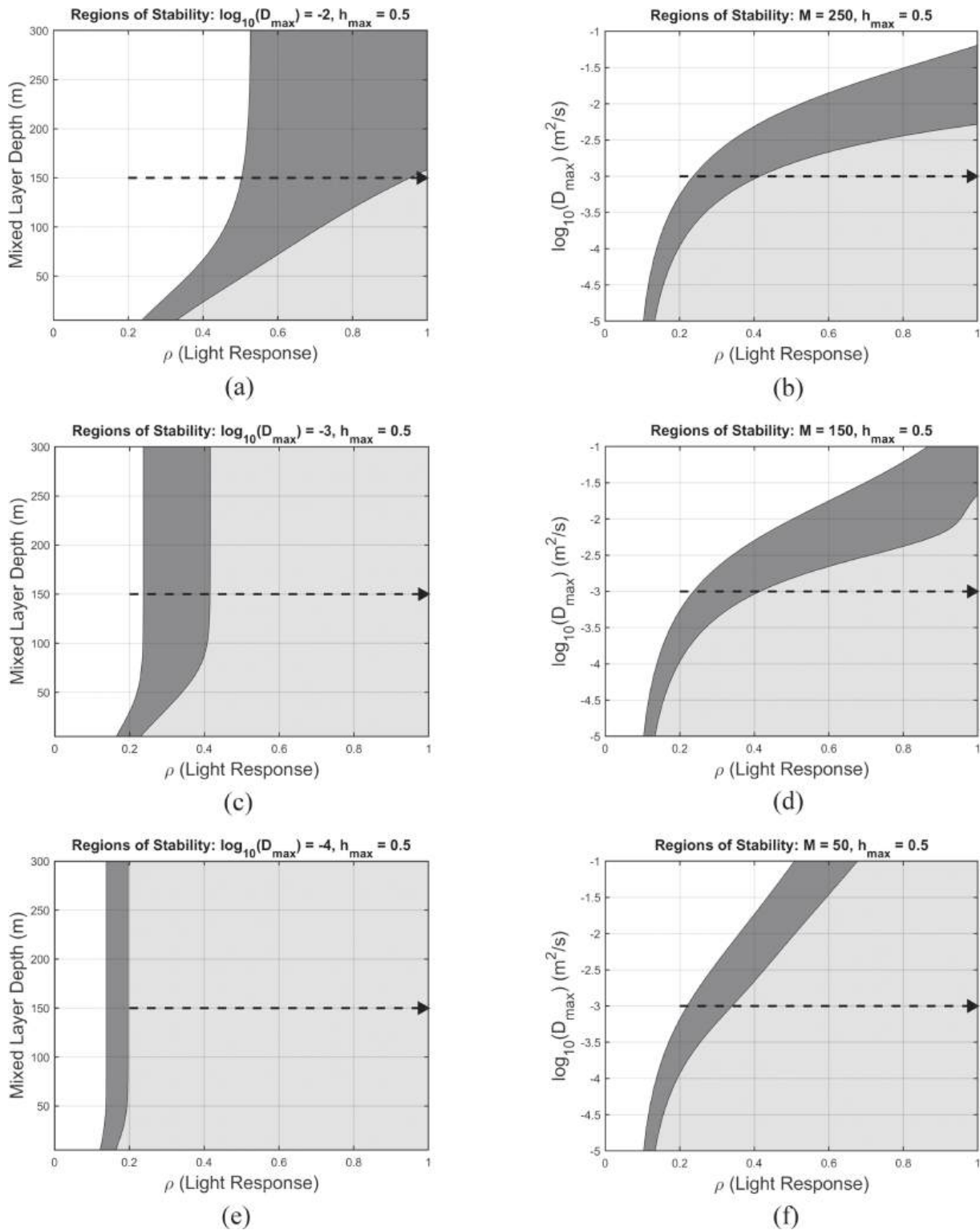


Fig. 7. Stability diagrams of equations (12) with parameter values from Tables I and II except where otherwise labeled. The diagrams are equivalent to those of Figure 5 except that $h_{\max}$ is reduced from 1.5 to 0.5 d^{-1} . In (a), (c) and (e), light response is plotted against mixed layer depth, and in (b), (d) and (f), light response is plotted against mixed layer diffusion. White region: plankton-free equilibrium is stable (left region); dark gray region: phytoplankton-only equilibrium is stable (middle region); light gray region: coexisting equilibrium is stable (right region). Dashed lines represent seasonal increases of light response, with the summer solstice at right arrow, for a certain level of mixed layer depth or mixed layer diffusion.

value of N_T would change accordingly. It is reasonable to expect that changing values of N_T , resulting from sinking phytoplankton and nutrient entrainment, would alter the equilibria of the reaction–diffusion model. However, based on the analysis of Busenberg *et al.* (Busenberg *et al.*, 1990) and the analysis of this paper, it is also reasonable to expect that the reaction–diffusion model will always have the same three equilibria (plankton-free, phytoplankton-only and coexisting), regardless of the value of N_T . It is also assumed in this paper that N_T is constant with depth, which is not so in the ocean. However, time-dependent depth profiles of N_T would be difficult to estimate and would complicate the process of solving the equations of the system. Nevertheless, valuable insights are gained from studying the reaction–diffusion system with temporally and spatially constant N_T .

CONCLUSION

In complex planktonic systems, it is not intuitive what the static plankton states would be if there is no seasonal forcing, or which parameters affect the mathematical attractors that influence the behavior of plankton populations. Light, diffusion and mixed layer depths have all been implicated as dominant drivers of plankton populations. In this paper, we determined the influence of fixed surface light, diffusion strength and mixed layer depth on the stability of the equilibria of an NPZ reaction–diffusion system. We envision our results giving ecologists and modelers theoretical expectations about what state the ecosystem ought to be under certain conditions of light, mixed layer depth and diffusion.

The seasonally changing quantities of diffusion and mixed layer depth have assumed roles in phytoplankton bloom theory as critical quantities that either restrict phytoplankton access to light (CDH), limit phytoplankton accumulation (CTH) or reduce contact rates between phytoplankton and zooplankton (DRH). The dynamical study of the model in this paper suggests that these two quantities, along with the impact of seasonally varying sunlight on growth rate, can cause a change in the attracting equilibrium state of the ecosystem. Behrenfeld and Boss (Behrenfeld and Boss, 2014) suggest the abiotic factors that cause blooms do so by causing imbalances in predator–prey relationships instead of by controlling growth rates of phytoplankton as theorized by the CDH and CTH. Our results suggest that seasonal changes in solar radiation, diffusion and, to a lesser degree, mixed layer depth can cause such ecological imbalances. Behrenfeld and Boss (Behrenfeld and Boss, 2014) also suggest that for the phytoplankton to bloom, they must first become scarce to reduce the grazing

population, and thus grazing pressure. This transition sets the stage for rapid phytoplankton growth due to weak grazing pressure. In our model, when sunlight intensity becomes too low and when diffusion (and in some cases, mixed layer thickness) becomes too high, the phytoplankton population declines and the attracting equilibrium of the model shifts to a phytoplankton-only state. Once the phytoplankton-only state becomes an attractor, zooplankton decline. Thus, contact rates between phytoplankton and zooplankton are reduced not only by dilution, as suggested by the DRH, but also by the switching of the stable state of the system. Our results suggest that this effect could occur during the transition into winter conditions of low sunlight, and high diffusion/deep mixed layer depth.

This study of a simple NPZ reaction–diffusion equation informs more complex plankton ecology models. Some plankton models have so many state variables or parameters that determining their equilibria and their stability properties can be difficult or impossible. However, we expect models that have similar structure and parameterizations to the model in this paper to exhibit similar dynamical behavior. Therefore, we anticipate that the results of this analysis will provide insight into future plankton modeling studies.

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